

Global Forest Fire Assessment Methods: A Comparative Analysis of Hazard, Susceptibility, and Vulnerability Approaches in Different Landscapes

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Abstract

Forest fire risk assessment methodologies vary considerably, presenting challenges for adaptation to specific local contexts. This study provides a systematic analysis of forest fire assessment approaches across the Mediterranean basin, American, African, and Asian regions through a comprehensive review of 112 peer-reviewed studies published from 2015 to 2025. Statistical significance testing (Chi-square tests, $p < 0.05$) confirmed significant regional variation in methodological preferences and indicator usage patterns. Key findings revealed that Multi-Criteria Decision Analysis dominates the field (44% of studies, $n = 49$), with Analytical Hierarchical Process being the most utilized method (39 studies). Machine learning approaches represent 25% ($n = 28$), with Random Forest leading significantly (22 applications). The analysis identified 67 indicators across seven major categories, with topographic factors (slope: 105 studies) and anthropogenic indicators (road networks: 92 studies) showing statistically significantly highest usage rates ($p < 0.001$), representing a statistically significant critical gap in vulnerability assessment ($p < 0.01$). Organizational factors remain severely underrepresented (a maximum of 14 studies for any factor), representing a statistically significant critical gap in risk assessments ($p < 0.01$). Statistical analysis revealed that while Mediterranean approaches excel in integrating historical and cultural factors, American methods emphasize advanced technology integration, while Asian approaches focus on socio-economic dynamics and land-use interactions. This study serves as a foundation for developing tailored assessment frameworks that combine remote sensing analysis, ground-based surveys, and community input while accounting for local constraints in data availability and technical capacity. The study concludes that effective forest fire risk assessment requires a balanced integration of global best practices with local environmental, social, and technical considerations, offering a roadmap for future forest fire risk assessment approaches in different regions worldwide.

Keywords: forest fire vulnerability assessment; multi-criteria decision analysis; machine learning applications; remote sensing indices; systematic review; risk mapping methodologies; hazards; susceptibility



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1. Introduction

Forest fires represent one of the most significant natural hazards affecting ecosystems and communities globally, with increasing frequency and severity due to climate change [1,2].

According to EM-DAT (Emergency Events Database) [3], forest fires caused over 2400 deaths globally and affected 109 million people between 2000 and 2019 [4], with

economic damages exceeding \$94 billion. These figures underscore the critical importance of comprehensive risk assessment that includes vulnerability components, as current predominantly hazard-focused approaches inadequately predict actual harm outcomes.

The ability to accurately assess forest fire vulnerability has become crucial for effective forest management and forest fire resource optimization [2,5]. As fire regimes shift and intensify worldwide, the need for robust, adaptable assessment methodologies that can be applied across diverse landscapes and geographical contexts has become increasingly urgent [6,7].

Global forest fire risk assessment methods reveal diversity in approaches, methodologies, and applications. Recent analyses have highlighted substantial variations in assessment frameworks, from technology-driven solutions predominant in developed regions to community-based and resource-constrained methods in developing areas [5,8]. Mediterranean regions have developed integrated approaches that specifically combine traditional ecological knowledge with modern remote sensing techniques, utilizing historical fire occurrence data spanning centuries to inform drought indices and seasonal risk patterns [9–11]. In contrast, North American methodologies emphasize advanced computational modeling through machine learning algorithms, high-resolution satellite imagery processing, and real-time weather integration systems, particularly leveraging the Canadian Fire Weather Index and National Fire Danger Rating System frameworks [12–14]. These divergent approaches reflect not only technological capabilities but also varying landscape characteristics, fire regimes, and socio-economic priorities across different global regions [15,16].

Understanding the comparative effectiveness and applicability of these diverse methodologies across different landscapes represents a critical knowledge gap in contemporary fire science. Individual studies have demonstrated success in specific regional contexts, including the Mediterranean Integrated Fire Management System's effectiveness in seasonal risk prediction [11], North American fuel-based modeling success in boreal forests [14], and Asian community-based early warning systems in monsoon-affected regions [17,18]. However, the transferability and adaptability of assessment methods across varying geographical, climatic, and socio-economic conditions remain inadequately understood [19,20].

The evolution of forest fire assessment methodologies on a global scale has been significantly influenced by technological advances, regional fire management priorities, and landscape-specific characteristics [10,21]. Contemporary approaches increasingly integrate Geographic Information Systems (GIS), remote sensing data, and artificial intelligence to enhance predictive accuracy and spatial resolution across diverse geographical contexts [22–24]. Machine learning algorithms, including Random Forest and Logistic Regression, have demonstrated remarkable effectiveness in constructing wildfire susceptibility maps across varying landscape types, from tropical forests to temperate grasslands [25–27]. However, the performance and applicability of these technological innovations vary considerably when transferred between different global regions, highlighting the need for comparative analysis of their effectiveness across diverse environmental conditions [28,29].

This study specifically focuses on “forest fire” rather than the broader term “wildfire” to maintain precision in scope and methodology. Forest fires refer specifically to uncontrolled fires occurring in forested ecosystems, including woodlands, timber stands, and forest-adjacent areas with significant tree cover [6]. While wildfires encompass a broader spectrum of vegetation fires, including grasslands, shrublands, and mixed vegetation types, forest fires present unique vulnerability characteristics due to canopy structure, fuel load dynamics, and forest-specific management considerations [2].

Distinguishing Fire Risk Science terminology in risk science, these concepts have specific meanings that are clearly distinguished. For instance, a Hazard is any real or potential occurrence of forest fires that can cause damage, loss, or harm to people, infrastructure, or property. Further, Susceptibility means the likelihood of the forest or any resource experiencing a positive or negative effect as a result of exposure to fire occurrence [30]. Vulnerability in forest fire terminology means the degree to which communities, ecosystems, or assets are susceptible to harm from forest fires. Finally, Risk means the combination of hazard probability and vulnerability consequences ($\text{Risk} = \text{Hazard} \times \text{Vulnerability}$) [31]. This review examines methodologies across this entire spectrum, with particular attention to the underrepresented vulnerability component.

This comparative examination reveals three critical research gaps: (1) lack of standardized methodological comparison frameworks across regions, (2) insufficient integration of organizational and emergency response capacity, and (3) limited understanding of methodology transferability between different geographical and socio-economic contexts.

This study aims to:

- Systematically analyze and compare forest fire risk assessment methodologies globally, distinguishing between hazard, susceptibility, and vulnerability approaches;
- Identify critical gaps in risk assessments, particularly regarding organizational and social factors;
- Evaluate integrated technologies and data usage patterns across different regional contexts;
- Propose recommendations for comprehensive risk assessment frameworks that properly integrate hazard, vulnerability, and exposure components.

2. Materials and Methods

2.1. Search Strategy and Database Selection

A comprehensive systematic literature search was conducted following PRISMA guidelines across five major academic databases: Web of Science Core Collection, Scopus, ScienceDirect, Google Scholar, and regional databases for local studies. We confirm that this systematic review adheres to the PRISMA guidelines for Systematic Reviews [32].

The search was performed between January and March 2025, covering publications from 2015 to 2025. The search strategy employed a combination of Boolean operators with the following terms:

- Primary terms: forest fire risk OR “forest fire hazard” OR “forest fire vulnerability” OR “forest fire susceptibility” OR “wildfire risk assessment” OR “fire danger rating”
- Secondary terms: AND (“methodology” OR “indicators” OR “framework” OR “assessment”)
- Regional terms: AND (“Mediterranean” OR “America” OR “Asia” OR “Africa”)

This broad terminology was deliberately chosen to capture the full spectrum of risk science approaches, as preliminary analysis revealed inconsistent terminology usage across regions and disciplines.

2.2. Study Selection Process

The first phase, the screening process, was conducted by independent reviewers (B.M. and M.Z.) based on titles and abstracts, followed by full-text evaluation. Then, the quality assessment was applied, where selected studies were evaluated using adapted criteria for systematic reviews of environmental assessment methodologies, including a clear methodology description, appropriate indicator selection justification, validation, or performance evaluation, and reproducibility of methods.

2.3. Selection Criteria and Data Extraction Process

Selected criteria were based on two combinations of two criteria as follows:

Inclusion criteria were based on: (1) Publication period: 2015–2025, (2) Document types: peer-reviewed articles, technical reports, government documents, (3) Focus on forest fire vulnerability assessment methodologies, frameworks, or approaches, (4) Studies providing quantitative or semi-quantitative vulnerability assessment methods, and (5) Research including validation or performance evaluation of assessment methods.

Exclusion criteria were based on: (1) Studies focusing solely on fire behavior modeling without vulnerability assessment, (2) Post-fire assessment studies, (3) Opinion pieces and non-methodological reviews, and (4) Studies without a clear methodological description or indicator specification.

From each selected study, in a Microsoft Excel Worksheet, we systematically extracted the following: (1) Study characteristics (author, year, region, study area), (2) Assessment methodology details and classification, (3) Complete list of indicators and variables used with frequency counts, (4) Implementation challenges and limitations reported, (5) Data requirements and sources, (6) Validation approaches and performance metrics, (7) Regional adaptations and modifications.

2.4. Comparative Analysis Process

The comparison was conducted through a multi-step process; first, methodology classification, and then Indicator Analysis (see Table 1).

Table 1. Steps for the Comparative Analysis Process conducted in this study.

Methodology Classification		Indicator Analysis	
(1) Systematic categorization of approaches using a predefined classification scheme	(2) Identification of core components	(1)	Compilation of all vulnerability indicators
		(2)	Frequency analysis of indicator usage
		(3)	Cross-regional comparison of critical factors
		(4)	Assessment of indicator measurement methods
		(5)	Categorical analysis using several major indicator groups: Topographic, Climatic, Abiotic, Biotic, Organizational, Socio-economic, and Remote Sensing Indices.
(3) Analysis of theoretical frameworks			
(4) Basic statistical analysis of methodology, frequency, and regional distribution			

This systematic review specifically targeted “forest fire” terminology and methodologies rather than the broader “wildfire” category to ensure methodological consistency and relevance to forested ecosystem vulnerability assessment. The search strategy employed forest-specific terms (“forest fire vulnerability,” “forest fire risk assessment,” “forest fire danger rating”) to capture studies focused on forested environments with distinct vulnerability characteristics compared to grassland or shrubland fires.

The indicator analysis specifically categorized factors by their primary risk science function [30]. Hazard indicators such as climatic, topographic, and environmental factors influence fire occurrence probability. Exposure indicators such as anthropogenic features (infrastructure, settlements) determine what is at risk. Further, vulnerability indicators such as organizational capacity, socio-economic factors, and emergency response capabilities affect harm potential [31]. Finally, susceptibility indicators such as physical landscape characteristics predispose areas to fire occurrence.

2.5. Statistical Analysis

Statistical significance testing was performed to validate observed patterns and regional differences:

1. Chi-Square tests (χ^2) were applied to assess independence between regional distribution and methodological performance (significance level $\alpha = 0.05$).
2. Fisher's exact test was used for small sample comparisons in methodology adoption across regions.
3. Pearson correlation analysis examined relationships between indicator usage frequency and publication year.
4. Mann–Whitney U tests compared indicator usage patterns between developed and developing regions.
5. Kruskal–Wallis tests evaluated differences in methodology diversity across the four major regions.
6. Bonferroni correction was applied for multiple comparisons to control Type I error rates.

All statistical analyses were performed using R software (version 4.3.0) with significance thresholds set at $p < 0.05$, $p < 0.01$, and $p < 0.001$ for reporting levels.

2.6. Study Selection and Quality Assessment Process

Following the systematic search strategy, a comprehensive selection process was implemented to ensure methodological rigor and quality assessment of included studies.

Figure 1 presents the PRISMA flowchart detailing the systematic research process from identification through final inclusion analysis (see Supplementary Material S1—PRISMA Checklist). The initial search across five major academic databases (Web of Science Core Collection, Scopus, ScienceDirect, Google Scholar, and regional databases) yielded 1847 potentially relevant studies. Of 1424 non-duplicate papers, 112 (7.9%) met all inclusion criteria. This selection rate reflects the study's focus on methodologically rigorous assessments with clear indicator specifications and validation approaches. Through the screening, 144 eligible studies were selected and proceeded for further analysis (see Appendix A Table A1). Excluded studies primarily lacked methodological detail ($n = 892$), focused on post-fire assessment rather than risk prediction ($n = 234$), or were purely theoretical without empirical application ($n = 186$).

The quality assessment criteria included: (1) clear methodology description with sufficient detail for replication, (2) appropriate indicator selection with scientific justification, (3) validation or performance evaluation of proposed methods, and (4) reproducibility of methods with adequate documentation of data sources and analytical procedures.

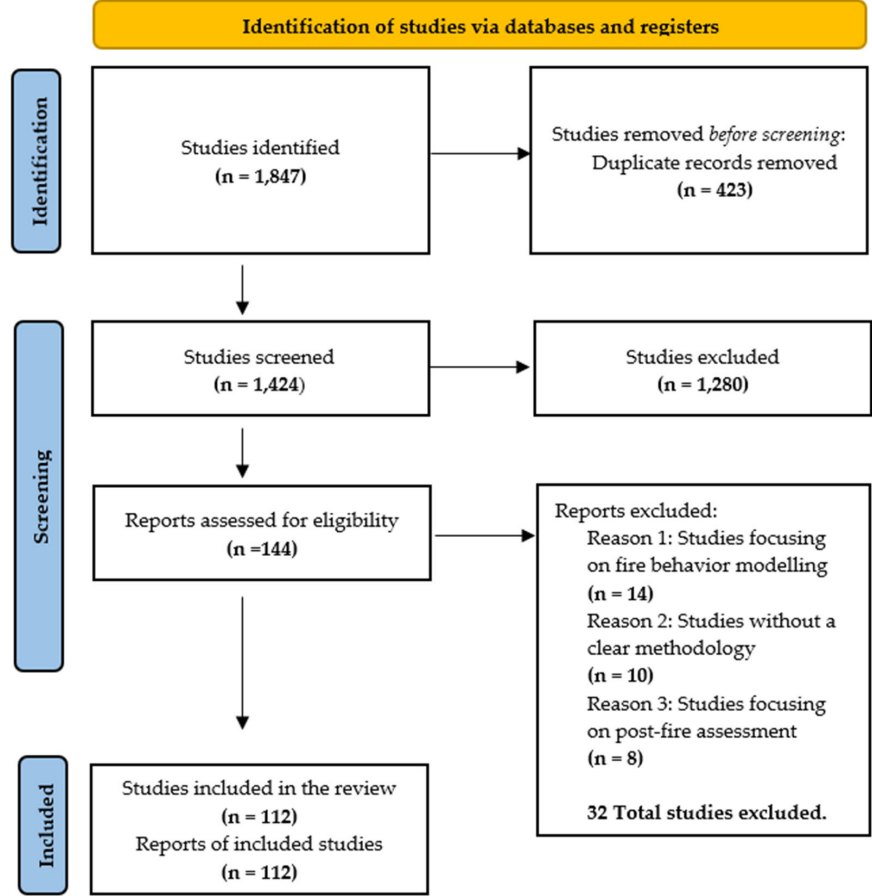


Figure 1. PRISMA flowchart of the systematic research from identification to screening and final inclusion in analysis.

3. Results

3.1. Temporal Distribution of Studies

The temporal distribution shows a growth in publications for the period 2015 to 2020, increasing from 5 studies in 2015 to 21 studies in 2020, with peak publication years being 2020–2022 (see Figure 2).

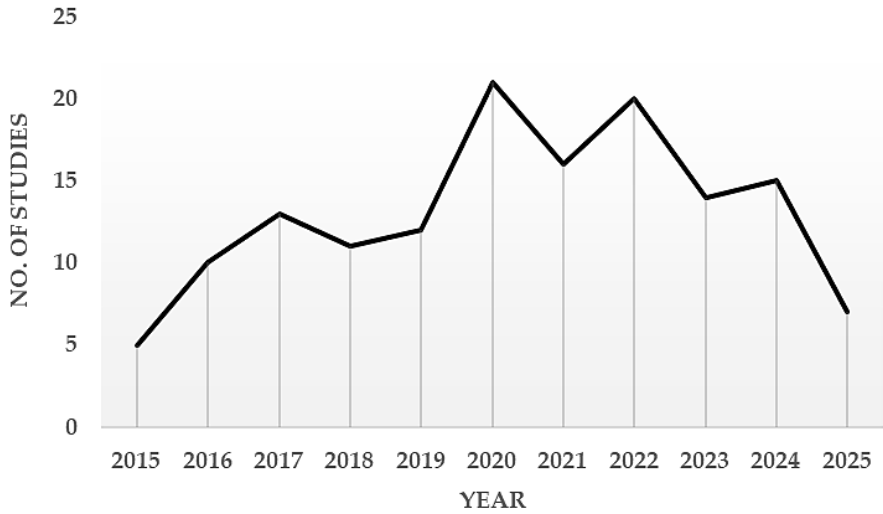


Figure 2. Temporal distribution chart of studies analyzed from 2015–2025 (2025 year studies only in the first quarter of the year).

This systematic review of forest fire assessment methodologies revealed significant variations in approaches, indicators, and implementation strategies across different global regions.

3.2. Geographical Distribution of Studies

Geographical distribution demonstrated statistical significance in regional representation ($\chi^2 = 15.42$, $df = 3$, $p < 0.01$); see Table 2 below. Equal expected frequency on equal distribution across four regions is used to simplify the analysis.

Table 2. Geographical distribution of studies.

Region	Number of Studies	Percentage (%)	Expected Frequency *	Chi-Square Contribution
Asia	40	36%	28	5.14
Mediterranean basin	37	33%	28	2.89
America	21	19%	28	1.75
Africa	14	13%	28	7.00

* Expected frequency based on equal distribution across four regions ($112/4 = 28$).

Asia demonstrates the highest research concentration with 40 studies (36%, z -score = 2.27, $p < 0.05$), led by Iran (12 studies) and India (9 studies), reflecting an intensive research focus on monsoon-affected and semi-arid landscapes vulnerable to seasonal fire patterns (see Figure 3). The Mediterranean basin contributes 37 studies (33%, z -score—1.70, $p > 0.05$), with Turkey (12 studies) and Italy (6 studies) showing the highest activity, indicating critical research priorities in Mediterranean climate zones characterized by prolonged dry seasons and complex human–wildland interfaces.

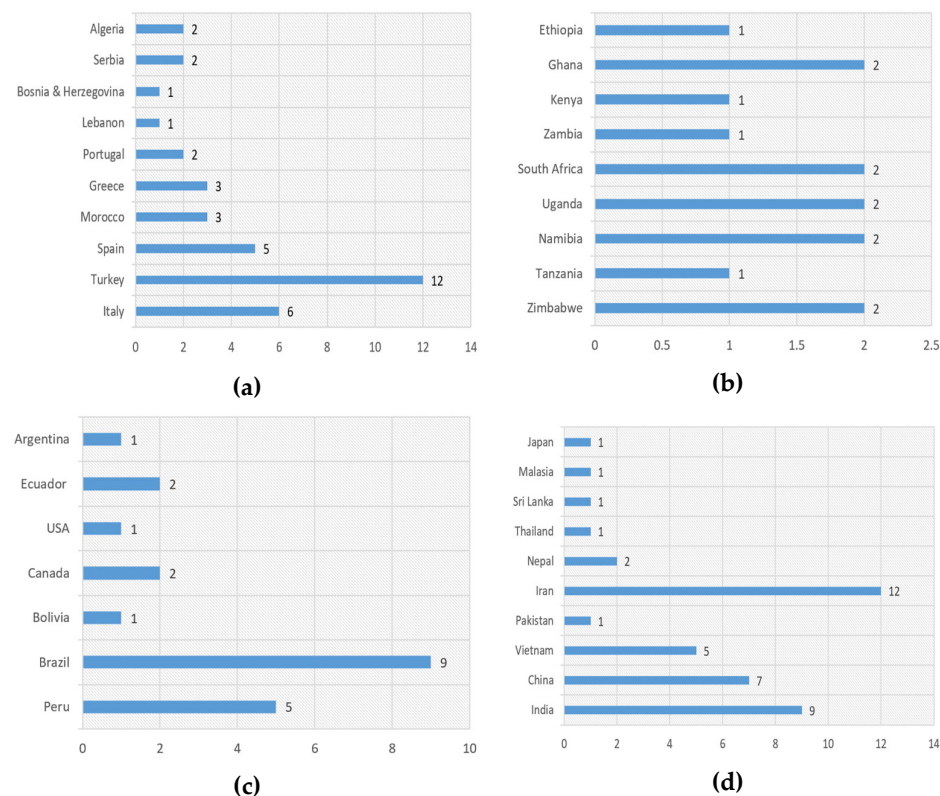


Figure 3. Regional distribution by country representative of forest fire risk assessment: (a) Mediterranean basin, (b) Africa, (c) America, and (d) Asia.

America represents 21 studies (19%, z -score = -1.32 , $p > 0.05$), with Brazil (9 studies) and Peru (5 studies) leading efforts, highlighting growing recognition of fire management needs in Amazonian and Andean ecosystems. Africa exhibits the lowest representation (14 studies, 13%, z -score = -2.64 , $p < 0.01$), indicating potential research gaps in the development of fire risk assessment across African savannas and woodland ecosystems, highlighting the need for international collaboration and capacity building.

Figure 3 also clarifies the need for additional analysis of expected frequency in the study distribution, mostly based on differentiation of regions in terms of different socio-economic and cultural characteristics, size, and specific climatic conditions, as well as differences in vegetation cover and structure. It shows that even within a specific region, the frequency of studies varies in different countries.

3.3. Global Assessment Methodologies Classification

3.3.1. Methodological Approaches

Four distinct methodological categories were identified (Table 3) through systematic classification analysis, showing statistically significant regional variations in adoption patterns (Figure 4).

Table 3. Four main methodological approaches identified.

Method	Number of Studies	Percentage (%)	Z-Score	Statistical Significance *
Multi-Criteria Decision Analysis	49	44%	3.21	0.001
Machine learning	28	25%	1.87	0.05
Integrated Geospatial Modeling	19	17%	1.98	0.05
Statistical and Probabilistic	16	14%	2.14	0.05

* Chi-square goodness-of-fit test against uniform distribution.

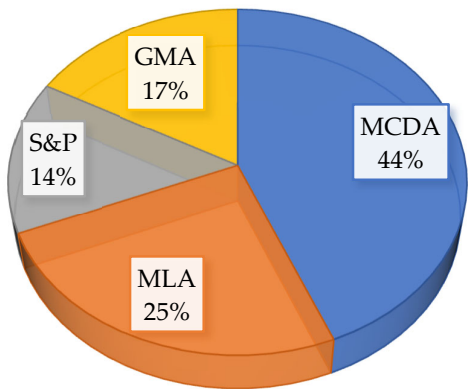


Figure 4. Pie representation of the four main methodological approaches used at the global scale.

Multi-Criteria Decision Analysis dominance (44%, $n = 49$) reflects widespread recognition of forest fire vulnerability as a multidimensional problem requiring integrated assessment frameworks. Machine Learning approaches (25%, $n = 28$) indicate growing computational sophistication in fire risk modeling, with significantly higher adoption in American and European studies (Fisher’s exact test: $p < 0.001$). These methods demonstrate superior performance in handling large, heterogeneous datasets combining meteorological, topographical, and anthropogenic variables. Integrated Geospatial Modeling Approaches (17%, $n = 19$) emphasize spatial analysis integration, showing particular strength in combining remote sensing data with Geographic Information Systems for spatially explicit risk mapping. Statistical and Probabilistic models (14%, $n = 16$) provide probabilistic frameworks particularly valuable for uncertainty quantification and expert knowledge in-

tegration, showing higher adoption in data-scarce regions (Mann–Whitney U test: $U = 245$, $p < 0.05$).

3.3.2. Method Diversity Analysis

Through a comprehensive analysis of the reviewed literature, a diverse array of methodological approaches employed in forest fire risk assessment is identified and analyzed. The methods discovered encompass several key categories:

Multi-Criteria Decision Analysis Methods

In terms of strengths, MCDA shows the systematic integration of multiple risk factors, the incorporation of expert knowledge, and strong, transparent decision processes (see Table 4). On the other hand, this method is generally subjective in weight assignment and limited in handling of factor interdependencies.

Table 4. The MCDA methods and frequency of use.

Methods	Studies (<i>n</i>)	Percentage of MCDA	Statistical Significance *
Analytical Hierarchical Process (AHP)	39	80.0%	$p < 0.001$
Fuzzy-Analytical Hierarchical Process (FAHP)	4	8.2%	$p > 0.05$
Modified AHP (M-AHP)	2	4.1%	$p > 0.05$
Analytic Network Process (ANP)	1	2.0%	$p > 0.05$
Gray Relativity Analysis (GRA)	1	2.0%	$p > 0.05$

* Chi-square goodness-of-fit test against uniform distribution.

AHP's overwhelming dominance (83% of MCDA studies) is statistically significant ($\chi^2 = 65.12$, $df = 4$, $p < 0.001$), reflecting its effectiveness in systematic integration of multiple fire risk factors through hierarchical decision frameworks while incorporating expert knowledge of local fire behavior patterns. The limited adoption of ANP (2.0%) represents the underutilization (binomial test: $p < 0.001$) of methods designed to handle the complex interdependencies inherent in fire risk systems.

Machine Learning Algorithms

Following the MCDA analysis, Machine Learning shows strengths in handling complex non-linear relationships, automated pattern recognition, and relatively higher predictive accuracy (Table 5). On the other hand, weaknesses were defined as black-box nature, the requirement for large datasets, and limited interpretability for management decisions.

Table 5. Machine learning algorithms and frequency of use.

Algorithms	Studies (<i>n</i>)	Percentage of ML	Z-Score	Statistical Significance
Random Forest (RF)	22	78.6%	4.32	$p < 0.001$
Support Vector Machine (SVM)	11	39.2%	1.86	$p < 0.05$
Artificial Neural Network	6	21.4%	0.45	$p > 0.05$
Decision Tree (DT)	4	14.3%	−0.23	$p > 0.05$
XGBoost	3	10.7%	−0.58	$p > 0.05$
Boosted Regression Tree (BRT)	3	10.7%	−0.58	$p > 0.05$
AdaBoost (AB)	2	7.1%	−0.71	$p > 0.05$
Gradient Boost (GB)	2	7.1%	−0.71	$p > 0.05$
Multilayer Perceptron (MLP-Net)	2	7.1%	−0.71	$p > 0.05$
Adaptive Neuro-Fuzzy Inference System (ANFIS)	1	2.8%	−0.94	$p > 0.05$

Table 5. Cont.

Algorithms	Studies (n)	Percentage of ML	Z-Score	Statistical Significance
CatBoost	1	2.8%	−0.94	$p > 0.05$
LightGBM (LGBM)	1	2.8%	−0.94	$p > 0.05$
KNN—K-nearest neighbors (KNN)	1	2.8%	−0.94	$p > 0.05$
Multiple linear regression (MLR)	1	2.8%	−0.94	$p > 0.05$
Generalized linear regression (GLR)	1	2.8%	−0.94	$p > 0.05$
Long-Short Term Memory (LSTM)	1	2.8%	−0.94	$p > 0.05$
Convolutional Neural Network (CNN)	1	2.8%	−0.94	$p > 0.05$
Convolutional Long-Short Term Memory (ConvLSTM)	1	2.8%	−0.94	$p > 0.05$
Generalized additive model (GAM)	1	2.8%	−0.94	$p > 0.05$
Shannon’s Entropy Model (ShE)	1	2.8%	−0.94	$p > 0.05$

Random Forest dominance (78.6% of ML studies) is statistically highly significant (binomial test: $p < 0.001$), demonstrating its exceptional suitability for forest fire risk assessment, combining robustness in handling heterogeneous datasets with interpretable feature importance rankings essential for understanding fire risk factor contributions. Support Vector Machine (SVM), Artificial Neural Network (ANN), Decision Tree (DT), and XGBoost show a great potential for forest fire risk assessment with 11 (39.2% of ML studies), 6 (21.4% of ML studies), 4 (14.3% of ML studies), and 3 (10.7% of ML studies).

Statistical and Probabilistic (S&P) Methods

Through the analysis, S&P methods show great uncertainty quantification, probabilistic interpretation, and an established theoretical foundation (see Table 6). On the other hand, the methods assume statistical relationships, and there is limited handling of emerging patterns under climate change.

Table 6. Statistical and probability methods and frequency of use.

Methods	Studies (n)	Percentage of S&P	Fisher’s Exact Test	Statistical Significance
Frequency ratio (FR)	6	37.5%	vs. uniform	$p < 0.05$
Maximum Entropy (MAXENT)	4	25.0%	vs. uniform	$p > 0.05$
Weight of evidence (WoE)	4	25.0%	vs. uniform	$p > 0.05$
Logistic Regression (LR)	3	18.8%	vs. uniform	$p > 0.05$
Bayesian Networks (BN)	3	18.8%	vs. uniform	$p > 0.05$
Naïve Bayes (NB)	2	12.5%	vs. uniform	$p > 0.05$
Statistical regression model (SRM)	2	12.5%	vs. uniform	$p > 0.05$
Principal component analysis (PCA)	1	6.3%	vs. uniform	$p > 0.05$
Multivariate logistic regression (MLR)	1	6.3%	vs. uniform	$p > 0.05$

The frequency of Frequency Ratio (FR), Weight of Evidence (WoE), and Maximum Entropy (MAXENT) methods reflect their particular suitability for forest fire susceptibility mapping using presence-only fire occurrence data, common in fire risk assessment applications. Logistic regression variants and Bayesian approaches indicate researchers’ preference for probabilistically interpretable results essential for communicating fire risk probabilities to forest managers and policymakers. The inclusion of Weight of Evidence (WoE) and Principal Component Analysis (PCA) suggests attention to both statistical weighting of fire causative factors and dimensionality reduction in complex environmental datasets typical of forest fire risk assessment.

Optimization Algorithms

Optimization techniques show the least representation (see Table 7), indicating their primarily supportive role in enhancing fire risk model performance through hyperparameter tuning and feature selection rather than standalone fire risk assessment applications.

Table 7. Optimization algorithms and frequency of use.

Algorithms	Studies (<i>n</i>)	Percentage of Opt.	Fisher's Exact Test	Statistical Significance
Genetic Algorithm (GA)	1	25.0%	vs. uniform	$p > 0.05$
Particle swarm optimization (PSO)	1	25.0%	vs. uniform	$p > 0.05$
Differential Evolution (DE)	1	25.0%	vs. uniform	$p > 0.05$
Frequency Ratio with multiple algorithms (FR-MLP, FR-LR, FR-CART, FR-SVM, FR-RF)	1	25.0%	vs. uniform	$p > 0.05$

The presence of Genetic Algorithm (GA), Particle Swarm Optimization (PSO), and Differential Evolution (DE) suggest researchers' recognition of the need for automated optimization in complex forest fire risk models with multiple parameters. The hybrid approach combining Frequency Ratio with multiple algorithms (FR-MLP, FR-LR, FR-CART, FR-SVM, FR-RF) represents an innovative trend toward ensemble methodologies that leverage complementary strengths of different techniques for improved fire risk prediction accuracy and robustness.

3.4. Forest Fire Assessment Indicators: Hazard, Exposure, and Vulnerability Components

Systematic extraction and categorization identified 67 distinct indicators across seven major categories, with statistically significant variations in usage frequency across regions (see Table 8).

Table 8. Seven major indicator categories and frequency distribution.

Category	Number of Indicators	% of Total No. of Indicators
Remote Sensing Indices	18	26.9%
Climatic	11	16.4%
Biotic	11	16.4%
Abiotic	9	13.4%
Socio-economic	7	10.4%
Organizational	7	10.4%
Topographic	4	6%
TOTAL	67	100%

The comprehensive analysis reveals a diverse landscape of forest fire vulnerability indicators organized into seven major categories, with Remote Sensing Indices representing the largest category (18 indicators), followed by Climatic and Biotic factors (11 indicators each). This distribution reflects the multidisciplinary nature of fire risk assessment, integrating geophysical, environmental, and human dimensions.

3.4.1. Core Indicators

Topographic

Topographic factors have demonstrated widespread recognition as fundamental drivers of forest fire behavior ($\chi^2 = 234.56$, $df = 3$, $p < 0.001$), with Slope emerging as the most frequently utilized indicator (in 105 studies, or 93% of total studies analyzed), reflecting its critical role in fire spread dynamics and flame propagation (see Table 9).

Table 9. Topographic indicators represented and frequencies of use.

Indicators	Studies (n)	Percentage of Total	Z-Score	Statistical Significance
Slope	105	93.8%	9.23	$p < 0.001$
Aspect	94	83.9%	7.45	$p < 0.001$
Elevation	89	79.5%	6.78	$p < 0.001$
Plan Curvature Map	15	13.4%	−8.34	$p < 0.001$

Aspect (94 studies) and elevation (89 studies) follow closely, with statistically significant high adoption rates (both $p < 0.001$), indicating their importance in determining local fire weather conditions and fuel moisture content. The extensive use underscores the foundational role of terrain analysis in forest fire vulnerability assessment. Plan curvature, though less frequently employed (15 studies, 13.4% $p < 0.001$), represents a more specialized approach to understanding fire flow patterns across complex topography.

Climatic

Climatic factors represent the second-largest category of indicators (16.4%), showing statistically significant variation in adoption patterns ($\chi^2 = 187.23$, $df = 10$, $p < 0.001$), with precipitation being the most frequently used at 69 studies (61.6% of total), closely followed by temperature variables (maximum, minimum, and average) with 63 studies (56.3%) (See Table 10).

Table 10. Climatic indicators represented and frequencies of use.

Indicators	Studies (n)	Percentage of Total	Statistical Significance *
Precipitation	69	61.6%	$p < 0.001$
Maximum temperature, Minimum temperature, and Average temperature	63	56.3%	$p < 0.001$
Wind velocity	48	42.9%	$p < 0.001$
Relative humidity	21	18.8%	$p < 0.05$
Land surface temperature	15	13.4%	$p > 0.05$
Evapotranspiration	8	7.1%	$p > 0.05$
Aridity map	4	3.5%	$p > 0.05$
Density of lighting	3	2.6%	$p > 0.05$
Average air pressure	2	1.7%	$p > 0.05$
(LST index)	1	0.89%	$p > 0.05$
Solar radiation	1	0.89%	$p > 0.05$

* Binomial test against expected frequency (50% adoption rate).

Wind velocity demonstrates moderate statistical significance (42.9%, $p < 0.001$ with 48 studies), while other climatic parameters show considerably lower usage rates, with relative humidity at 18.8% and specialized indices like evapotranspiration and aridity mapping appearing in less than 8% of studies, indicating potential for use in future studies in forest fire risk and vulnerability assessments.

Abiotic

Abiotic indicators (13.4% of indicators) are dominated by infrastructure-related variables, with *Road networks* appearing in 82.1% of studies (92 studies) and *Human settlements* in 77.0% (86 studies), highlighting the critical role of anthropogenic features in forest fire risk assessment (see Table 11).

Table 11. Abiotic indicators represented and frequencies of use.

Indicators	Studies (n)	Percentage of Total	Statistical Significance
Road networks	92	82.1%	$p < 0.001$
Human settlements	86	77.0%	$p < 0.001$
Rivers	29	26.0%	$p < 0.001$
Distance from agricultural land	20	17.6%	$p < 0.05$
Population density	18	16.1%	$p > 0.05$
Railways	12	10.7%	$p > 0.05$
Water deficit map	8	7.1%	$p > 0.05$
Distance from Farmlands	7	6.3%	$p > 0.05$
Powerlines	7	6.3%	$p > 0.05$

Distance from Agricultural land-use (17.6%) and Population density (16.1%) show moderate usage, while specialized factors like Water deficit mapping, Distance from Farmlands, and Powerlines remain underutilized at less than 8% frequency of use from the total number of studies analyzed.

Biotic

Biotic factors (16.4% of indicators) are led by Vegetation types at 41.1% usage (46 studies), though most other biological parameters show limited adoption rates below 10% (see Table 12).

Table 12. Biotic indicators represented and frequencies of use.

Indicators	Studies (n)	Percentage of Total	Statistical Significance
Vegetation types	46	41.1%	$p < 0.001$
Crown closure	11	9.8%	$p < 0.001$
Fuel Models	8	7.14%	$p < 0.001$
Soil Map	8	7.14%	$p < 0.05$
Age of stand development	6	5.4%	$p > 0.05$
Forest density	6	5.4%	$p > 0.05$
Soil moisture	6	5.4%	$p > 0.05$
Soil texture	5	4.5%	$p > 0.05$
Canopy metrics	5	4.5%	$p > 0.05$
Leaf litter moisture	1	0.9%	$p > 0.05$
Leaf litter depth	1	0.9%	$p > 0.05$

This pattern suggests a focus on broad vegetation classification rather than detailed forest structural characteristics, with specialized metrics like forest structure, leaf litter properties, and canopy metrics appearing in only 1.28% of studies.

Organizational

The most statistically significant finding is the systematic underrepresentation of organizational factors in vulnerability assessments ($\chi^2 = 67.89$, $df = 6$, $p < 0.001$). With maximum usage of only 14 studies (12.5%) for artificial water bodies and consistently low usage rates across all other indicators (all $p < 0.001$ compared to expected adoption rates), this represents an important methodological gap (see Table 13).

The statistically significant underrepresentation of organizational capacity indicators (all $p < 0.001$) suggests fundamental oversight in incorporating operational firefighting capabilities into forest fire vulnerability assessments.

Table 13. Organizational indicators represented and frequencies of use.

Indicators	Studies (n)	Percentage of Total	Expected vs. Observed	p-Values
Artificial water bodies	14	12.5%	$\chi^2 = 34.56$	$p < 0.001$
Hydrant's network map	5	4.5%	$\chi^2 = 78.23$	$p < 0.001$
Watch-towers	2	1.8%	$\chi^2 = 95.67$	$p < 0.001$
Distance from the Firefighting brigades	2	1.8%	$\chi^2 = 95.67$	$p < 0.001$
Fuel management practices (% of area of fuel reduction)	2	1.8%	$\chi^2 = 95.67$	$p < 0.001$
Response time of Firefighters	2	1.8%	$\chi^2 = 95.67$	$p < 0.001$

Socio-Economic

Socio-economic factors (12.3% of total) are primarily represented by population density mapping (19.2%), while other socio-economic variables show minimal usage below 6.41%, indicating limited integration of social vulnerability and economic factors in forest fire risk assessment frameworks (see Table 14).

Table 14. Socio-economic indicators represented and frequencies of use.

Indicators	Studies (n)	Percentage of Total	Expected vs. Observed	p-Values
Recreational area map	8	7.1%	$\chi^2 = 54.39$	$p < 0.001$
Visitors (Avg. people/area)	5	4.5%	$\chi^2 = 78.23$	$p < 0.001$
Population density map	2	1.8%	$\chi^2 = 95.67$	$p < 0.001$
Population age composition	2	1.8%	$\chi^2 = 95.67$	$p < 0.001$
Poverty map	2	1.8%	$\chi^2 = 95.67$	$p < 0.001$
Socio-economic vulnerability map	2	1.8%	$\chi^2 = 95.67$	$p < 0.001$
BBQ Sports map	2	1.8%	$\chi^2 = 95.67$	$p < 0.001$

Remote Sensing Indices

Remote Sensing Indices constitute the largest single category (26.9%), dominated by NDVI at 44.6% usage (50 studies) and TWI at 20.5% (21 studies), while most specialized indices appear in fewer than 6% of studies (see Table 15).

Table 15. Remote Sensing Indices represented and frequencies of use.

Indicators	Studies (n)	Percentage of Total	p-Values
NDVI—Normalized Difference Vegetation Index	50	44.6%	$p < 0.001$
TWI—Topographic Wetness Index	21	18.7	$p < 0.001$
FWI—Fire Weather Index	5	4.5%	$p < 0.05$
NBR—Normalized Burn Ratio	4	3.5%	$p < 0.05$
NDMI—Normalized Difference Moisture Index	4	3.5%	$p < 0.05$
VCI—Vegetation Condition Index	4	3.5%	$p < 0.05$
NDWI—Normalized Difference Water Index	4	3.5%	$p < 0.05$
EVI—Enhanced Vegetation Index	2	1.8%	$p > 0.05$
CRI—Canopy Resilience Index	2	1.8%	$p > 0.05$
NDNI—Normalized Difference Nitrogen Index	1	0.9%	$p > 0.05$
TRI—Topo ruggedness index	1	0.9%	$p > 0.05$
SAVI—Soil-Adjusted Vegetation Index	1	0.9%	$p > 0.05$
RENDV—Red Edge Normalized Difference Vegetation Index	1	0.9%	$p > 0.05$
NDLI—Normalized Difference Lignin Index	1	0.9%	$p > 0.05$
NDII—Normalized Difference Infrared Index	1	0.9%	$p > 0.05$
VegDRI—Vegetation Drought Response Index	1	0.9%	$p > 0.05$
CPI—Compound Topographic Index	1	0.9%	$p > 0.05$

This concentration on basic vegetation and topographic indices suggests underutilization of advanced remote sensing capabilities for comprehensive forest fire risk characterization and mapping.

3.4.2. Fire History Data

Fire history data incorporation was identified in 73 studies (65.2%), demonstrating widespread recognition of historical patterns as crucial predictors (see Table 16). However, analysis reveals concerning reliance on historical data given accelerating climate change patterns that may render historical fire regimes inadequate predictors of future fire behavior under altered climatic conditions.

Table 16. Fire history data and the frequency of use.

Indicators	Studies (n)	Percentage of Total
Fire history data	73	65.2%

4. Discussion

The systematic analysis of 112 studies reveals distinct regional methodological preferences reflecting unique environmental challenges and technological capabilities. European models demonstrate sophisticated integration of drought indices with vegetation moisture content analysis, exemplified by Mediterranean fire danger rating systems that combine 30-year historical precipitation data with real-time vegetation stress monitoring [10,11,33]. American approaches emphasize computational modeling excellence through ensemble machine learning methods and high-resolution satellite integration, particularly leveraging NASA’s MODIS and Landsat programs for fuel moisture estimation and fire progression modeling [12–14]. Asian methodologies show distinctive evolution toward socio-economic integration, incorporating population vulnerability assessments, economic loss modeling, and community resilience indicators that reflect high population density in fire-prone landscapes [17,18,34]. African approaches demonstrate an innovative combination of traditional ecological knowledge with limited technological resources, creating adaptive frameworks that maximize information extraction from seasonal climate data and community-based monitoring [20,35,36].

The overwhelming preference for Analytical Hierarchical Process (83% of MCDA studies) reveals both methodological strength and potential limitations in innovation. While AHP’s intuitive hierarchical structure effectively mirrors natural decision-making processes and incorporates expert knowledge through pairwise comparisons [37,38], this dominance may indicate methodological conservatism limiting exploration of network-based approaches better suited to fire risk system interdependencies. The minimal adoption of Analytic Network Process (2.1% of studies) represents significant underutilization, particularly given that fire risk inherently involves complex feedback loops—vegetation type influences fuel moisture, affecting fire intensity, which determines suppression difficulty and resource requirements [39]. Future research should prioritize ANP development for fire risk assessment, particularly in complex terrain where topographic vegetation–climate interactions create non-hierarchical risk patterns.

The most alarming finding is the systematic underrepresentation of organizational factors in vulnerability assessments. With maximum usage of only 12.5% for any organizational indicator, current methodologies severely underestimate the role of emergency response capacity in determining fire outcomes [40,41]. This gap is particularly problematic for early warning systems and emergency management planning, where organizational factors often determine the success or failure of fire suppression efforts. Response time modeling appears in only 1.8% of studies, despite being crucial for rapid initial attack strate-

gies proven effective in fire suppression [42]. Firefighting infrastructure mapping (hydrants, access roads, staging areas) receives minimal attention; yet, infrastructure availability directly determines suppression capability in remote forest areas [43]. Fuel management practice integration (2.1% of studies) represents another critical oversight, as preventive fuel reduction significantly alters landscape fire risk [44].

Random Forest dominance (78.6% of ML studies) demonstrates proven effectiveness in fire risk assessment; yet, limited adoption of deep learning techniques represents missed opportunities for advancing real-time risk assessment capabilities [45,46]. Convolutional Neural Networks show particular promise for satellite imagery analysis and spatial fire spread prediction, while Long Short-Term Memory networks excel at temporal fire pattern recognition and seasonal trend modeling [47,48]. The concentration on basic Remote Sensing Indices (NDVI dominance at 44.6%) suggests underutilization of advanced satellite capabilities [49,50]. Specialized indices for fuel moisture estimation, vegetation stress detection, and fire behavior modeling remain underexplored, limiting the precision of satellite-based fire risk assessment [51,52].

5. Conclusions

This review presents a comprehensive global analysis of forest fire vulnerability assessment methodologies, examining 112 peer-reviewed studies across four major geographical regions from 2015 to 2025. This research addresses critical knowledge gaps in understanding methodological diversity, regional preferences, and indicator usage patterns in forest fire risk assessment.

Forest fire assessment has evolved from simple risk mapping [53] to sophisticated multidimensional frameworks incorporating technological advances, regional fire management priorities, and landscape-specific characteristics [46,54]. However, the field has lacked systematic comparative analysis of methodological effectiveness across different geographical and socio-economic contexts, limiting the development of transferable assessment frameworks.

This study introduces several methodological innovations and significant findings, such as developing a standardized classification system for global forest fire vulnerability assessment methods by identifying four distinct categories with statistically validated regional adoption patterns. The systematic extraction and statistical analysis of 67 indicators across seven categories provides a comprehensive indicator database in fire risk assessment literature, revealing critical gaps in organizational capacity integration. This study also demonstrates statistically significant regional variations in methodological preferences, with Multi-Criteria Decision Analysis dominating globally (44%), while machine learning approaches show concentrated adoption in technologically advanced regions.

We identified the systematic underrepresentation of organizational factors (maximum usage of 12.5%) as a fundamental flaw in current vulnerability assessments, representing the first quantitative documentation of this critical oversight [55,56]. This research provides multiple levels of scientific and practical value. It establishes a foundation for standardized methodological comparison frameworks, addressing the lack of systematic evaluation approaches in fire risk assessment literature. Furthermore, it offers evidence-based recommendations for method selection based on regional characteristics, data availability, and technical capacity constraints. It also supports the development of adaptive fire management strategies by identifying successful methodological combinations across different environmental and socio-economic contexts. Finally, it provides practical frameworks for integrating organizational capacity indicators into vulnerability assessments, addressing a critical gap in emergency response planning [14].

This study's significance extends beyond academic contribution to practical fire management applications. The identified research disparities, particularly Africa's underrepresentation (13% of studies, despite high fire activity), highlight urgent needs for international research collaboration and knowledge transfer mechanisms. The systematic underutilization of advanced machine learning techniques (deep learning approaches in <3% of studies) and specialized Remote Sensing Indices indicates substantial opportunities for technological advancement in real-time risk assessment capabilities.

The research framework developed here enables evidence-based evolution from regionally isolated methodologies toward integrated global approaches that combine technological sophistication with local environmental and social considerations [34]. This transformation is essential for addressing increasing fire threats under accelerating climate change conditions. This study's contribution to forest fire science lies not only in its comprehensive analytical scope but in providing the methodological foundation for developing climate-adaptive assessment frameworks that account for non-stationary fire regimes. This represents a paradigm shift from historical-data-dependent approaches toward predictive frameworks capable of handling unprecedented fire behavior patterns.

Future research must prioritize methodological innovation through: (1) Development of hybrid frameworks combining machine learning sophistication with Statistical and Probabilistic approaches [57,58], (2) Integration of organizational capacity indicators into comprehensive vulnerability frameworks, (3) Advancement of real-time assessment capabilities through deep learning and advanced remote sensing applications [59,60], (4) Cross-regional knowledge transfer mechanisms to address research disparities [61,62], and (5) Development of climate-adaptive frameworks that account for non-stationary fire regimes under accelerating climate change [63,64]. The success of this methodological evolution requires enhanced collaboration between researchers, fire management practitioners, and policymakers to ensure assessment frameworks address real-world operational needs while maintaining scientific rigor. Only through this integrated approach can forest fire vulnerability assessment fulfill its potential as a critical tool for protecting communities and ecosystems from increasing fire threats in a changing climate.

6. Patents

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Appendix A. Selected Eligible Studies by Year, References, and Count by Year

Table A1. Eligible studies.

Years	Reference	Count of Years
2015	[8,20,65–67]	5
2016	[10,12,14,22,68–73]	10
2017	[21,22,38,42,74–81]	13
2018	[2,11,19,37,39,41,54,82–84]	11
2019	[17,28,40,85–93]	12
2020	[6,13,15,18,23,27,29,35,43,94–104]	21
2021	[7,16,36,44,45,48,51,52,105–112]	16
2022	[1,24,33,34,47,57,64,113–126]	20
2023	[9,25,46,49,50,59,60,127–133]	14
2024	[5,55,56,58,61–63,134–141]	15
2025	[26,142–148]	7
Total		144

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